

Lecture 7: American options, Continuous-time Conditional Expectation and Martingales

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1 Optimal Exercise of American Options

We have seen in Lecture 6 that the optimal exercise time for an American option is a stopping time τ^* determined by

$$\tau^* = \min\{k, V_k = G_k\}. \quad (1)$$

That is, the optimal exercise time is the first time that the value of the American option is equal to its intrinsic value. The reason why this τ^* is optimal is because, the discounted value process of the option, stopped at τ^* , is a martingale¹. As a result,

$$\begin{aligned} V_0 &= \tilde{E}\left(\frac{V_{N \wedge \tau^*}}{(1+r)^{N \wedge \tau^*}}\right) = \tilde{E}\left(\mathbb{I}_{\{\tau^* \leq N\}} \frac{V_{\tau^*}}{(1+r)^{\tau^*}}\right) + \tilde{E}\left(\mathbb{I}_{\{\tau^* = \infty\}} \frac{V_N}{(1+r)^N}\right) \\ &= \tilde{E}\left(\mathbb{I}_{\{\tau^* \leq N\}} \frac{G_{\tau^*}}{(1+r)^{\tau^*}}\right) + 0 = \tilde{E}\left(\mathbb{I}_{\{\tau^* \leq N\}} \frac{G_{\tau^*}}{(1+r)^{\tau^*}}\right). \end{aligned} \quad (2)$$

Thus, the price at time zero is the expectation of discounted payoff if we exercise the option at time τ^* .

Remark 1. From the above proof we see that the optimality of τ^* lie in the fact that τ^* can turn the discounted value process into a martingale. This is still the case when we consider continuous-time models.

Remark 2. For an European option, we can use a portfolio of stocks and cash to replicate its payoff at maturity N . For an American option, since it is optimal to exercise it at τ^* , we can similarly setup a portfolio of stocks and cash at time zero, rebalance the stock position at every time step. This portfolio will be held until time τ^* , when the American option is optimally exercised. A formula to express this strategy is:

$$\text{Option} = \text{Stocks} + \text{Cash earning risk-free rate, until the option is (optimally) exercised.} \quad (3)$$

Similarly, we can construct a portfolio of an European option and stocks, rebalance the position of stocks so that there is no risk associated with this portfolio. If we hold this portfolio until the maturity N , the value of the portfolio will be exactly equal to $(1+r)^N$ times the initial cost of the portfolio at time zero. That is, if we borrowed X dollars to buy the European option, then we can receive $(1+r)^N X$ from the portfolio at time N to pay off our loan. For American options, the story is the same². If we borrowed X dollars to buy an American option, then we can trade in stocks and cash, so that the portfolio consisting of this American option, stocks and cash has no risk at any time. The portfolio is held until time τ^* , when we exercise the option optimally. Then we can still receive $(1+r)^{\tau^*} X$ from this portfolio at that time, and use it to pay off our loan. A formula to express this strategy is:

$$\text{Cash earning risk-free rate} = \text{Option} - \text{Stocks, until the option is (optimally) exercised.} \quad (4)$$

¹The detailed proof is given in Theorem 4.4.5 of the textbook. It is really a verification of martingale property.

²For details, check Exercise 4.2 of Vol I.

2 General Probability Space and Random Variables

Let $(\Omega, \mathcal{F}, P)$ be probability space with probability measure P . A random variable X on this space is a real-valued (measurable) function on Ω . Random variables can be seen as realizations of the underlying probability space. For example, we know the results of coin tosses if we see the paths of stock prices. More specifically, given a random variable X , one may define a probability measure P_X on the real line: for any $x \in \mathbb{R}$, define

$$F_X(x) := P_X(X \leq x) = P(\omega; X(\omega) \leq x). \quad (5)$$

The function F_X is called the cumulative distribution function (CDF) of the random variable X . Because of (3), it is usually convenient and sufficient to just specify F_X . The CDF F_X is obviously monotonically increasing. And if there exist a non-negative function f such that

$$F_X(x) = \int_{-\infty}^x f(t)dt, \text{ for all } x, \quad (6)$$

then f is called the probability density function (PDF) of the random variable X . We assume the random variable X has a PDF henceforth.

Example 1. A normal random variable has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (7)$$

In probability theory, it is usually useful to compute some statistics of a given random variable. The mean, or the expectation of a random variable can be computed as

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx. \quad (8)$$

The variance of X , is defined in terms of the deviation of X from its mean:

$$\text{Var}(x) = E(X - E(X))^2 = E(X^2) - (E(X))^2. \quad (9)$$

The moment-generating function of X is defined as

$$E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)dx. \quad (10)$$

For the normal random variable in Example 1, the mean, the variance and the moment-generating function are respectively μ , σ^2 and $e^{\mu t + \frac{1}{2}\sigma^2 t^2}$.

Given a random variable X , one can compute the conditional expectation of X . If the σ -algebra $\mathcal{F}$ is given, then everything is known, X is not a random number any more, so we have

$$E(X|\mathcal{F}) = X.$$

If we are given the trivial σ -algebra $\mathcal{F}_0 = \{\phi, \Omega\}$, then we still know nothing about X . So the best estimation of X would be its expectation

$$E(X|\mathcal{F}_0) = E(X).$$

If we are given a non-trivial σ -algebra of $\mathcal{F}$: $\mathcal{G} \subset \mathcal{F}$, then the conditional expectation is no longer trivial. The conditional expectation, $E(X|\mathcal{G})$, can be interpreted the best estimation³ of X given the information $\mathcal{G}$. The information $\mathcal{G}$ could be irrelevant to X , in this case we say that X is independent of $\mathcal{G}$, and the conditional expectation of X based on $\mathcal{G}$ is simply the expectation of X .

Example 2. Consider two jointly normal random variables with the joint density

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right\}. \quad (11)$$

³See Exercise 2.7 in the textbook.

That is, X and Y are two standard normal random variables. They have a correlation of ρ , so knowing values of X will help estimating Y . Thus, one can finish the (tedious) computation using conditional probability density

$$E(Y|X) = \int_{-\infty}^{\infty} y \cdot \frac{f_{X,Y}(x,y)}{f_X(x)} dy, \text{ where } f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}. \quad (12)$$

Alternatively, we can use the fact that X and Y have a correlation of ρ , and use a orthogonalization procedure to rewrite Y :

$$Y = \rho X + \sqrt{1 - \rho^2} Z, \quad (13)$$

where Z is a standard normal independent of X . Then the conditional expectation of Y given X is simply

$$E(Y|X) = \rho X + E(Z|X) = \rho X + E(Z) = \rho X. \quad (14)$$

General conditional expectations have the same property as the discrete conditional expectation we learned in N -period binomial model.

- Taking out what is known

$$E(XY|\mathcal{G}) = XE(Y|\mathcal{G}), \text{ if } X \text{ is } \mathcal{G}\text{-measurable.}$$

- Iterated conditioning: if $\mathcal{H} \subset \mathcal{G}$, then

$$E(E(X|\mathcal{G})|\mathcal{H}) = E(X|\mathcal{H}).$$

In particular,

$$E(E(X|\mathcal{G})) = E(X).$$

- Independence: if X is independent of $\mathcal{G}$, then

$$E(X|\mathcal{G}) = E(X).$$

3 Continuous-time Stochastic Processes and Martingales

On the probability space $(\Omega, \mathcal{F}, P)$, let $T < \infty$ be a time-horizon. Assume that $\bar{\mathcal{F}} = \{\mathcal{F}_t\}_{t \in [0, T]}$ is a filtration on this probability space. Thus, $\mathcal{F}_s \subset \mathcal{F}_t$ for all s and t such that $0 \leq s \leq t \leq T$. The set $\mathcal{F}_t$ can be interpreted as available information at time t . A claim holds almost surely, “a.s.”, if it holds on a set $A \in \mathcal{F}$ such that $P(A) = 1$.

A stochastic process $X = \{X_t\}_{t \in [0, T]}$ is said to be adapted to the filtration $\bar{\mathcal{F}}$ if X_t is an $\mathcal{F}_t$ -measurable random variable for all $t \in [0, T]$. This means at time t , we are able to know the value of X_t with certainty. A stochastic process $\{X_t\}_{t \in [0, T]}$ is said to be continuous if for any $\omega \in \Omega$, $X_t(\omega)$ is a continuous real-valued function of $t \in [0, T]$ - in other words if all paths of $\{X_t\}_{t \in [0, T]}$ are continuous.

A stochastic process $M = \{M_t\}_{t \in [0, T]}$ is called an $(\bar{\mathcal{F}}, P)$ -martingale if:

- M is $\bar{\mathcal{F}}$ -adapted,
- $E(|M_t|) < \infty$ for all $t \in [0, T]$,
- $E(M_t|\mathcal{F}_s) = M_s$ for all s and t such that $0 \leq s \leq t \leq T$.

Example 3. Let X_k be i.i.d. random variables that take ± 1 :

$$X_k = \begin{cases} 1 & \text{with probability 0.5} \\ -1 & \text{with probability 0.5} \end{cases} \quad (15)$$

Define process $M_0 = 0$,

$$M_k = \sum_{j=1}^k X_j, \quad j = 1, 2, \dots \quad (16)$$

The process $\{M_k\}$ is called a symmetric random walk. It is a martingale. Moreover,

- $M_m - M_n$ is independent of M_k if $k \leq n \leq m$;
- The quadratic variation: $\sum_{j=1}^k (M_j - M_{j-1})^2 = \sum_{j=1}^k 1 = k$;
- The first order variation: $\sum_{j=1}^k |M_j - M_{j-1}| = \sum_{j=1}^k 1 = k$;

Now fix a $T > 0$ and large positive integer N , and define

$$X_t = \sqrt{\frac{T}{N}} M_{N \frac{t}{T}}, \quad t = 0, \frac{T}{N}, \frac{2T}{N}, \dots, T, \quad (17)$$

and use linear interpolation to define X_t for general t . Then we again $\{X_{\frac{kT}{N}}\}$ is a martingale, and

- $X_{\frac{mT}{N}} - X_{\frac{nT}{N}}$ is independent of $X_{\frac{kT}{N}}$ if $k \leq n \leq m \leq N$;
- The quadratic variation: $\sum_{j=1}^N (X_{\frac{jT}{N}} - X_{\frac{(j-1)T}{N}})^2 = \frac{T}{N} \sum_{j=1}^N 1 = T$;
- But The first order variation: $\sum_{j=1}^N |X_{\frac{jT}{N}} - X_{\frac{(j-1)T}{N}}|^2 = \sqrt{\frac{T}{N}} \sum_{j=1}^N 1 = \sqrt{NT}$;